

CASE STUDY: BRAYS BAYOU FLOOD FREQUENCY

An aerial photograph showing a residential neighborhood completely inundated with floodwater. The water is a murky, brownish-grey color, covering the streets, yards, and surrounding areas. Numerous houses with various roof colors (brown, grey, red) are visible, partially submerged. Green trees are scattered throughout the landscape, some standing in the water. A winding road or path is visible on the right side of the image, also surrounded by water. The overall scene depicts a severe flooding event.

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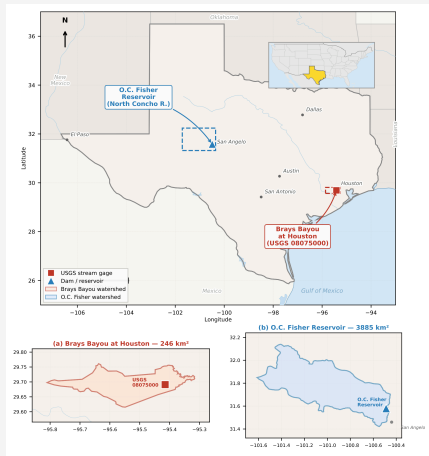
TODAY

We discuss Smith et al. (2026), a nonstationary flood frequency analysis of Brays Bayou at Houston, fit by Bayesian methods in the open-source RMC-BestFit software from the US Army Corps of Engineers.

! Important

Every figure is from the paper, which is open access under CC BY 4.0, so it's OK to share here.

CASE STUDY: TWO TEXAS WATERSHEDS



Brays Bayou and O.C. Fisher Reservoir (Smith et al., 2026, Fig. 1, CC BY 4.0).

Clear physical drivers and strong nonstationary signals

- **Brays Bayou:** floods rising with urbanization
- **O.C. Fisher:** floods falling with brush encroachment and groundwater pumping

The data

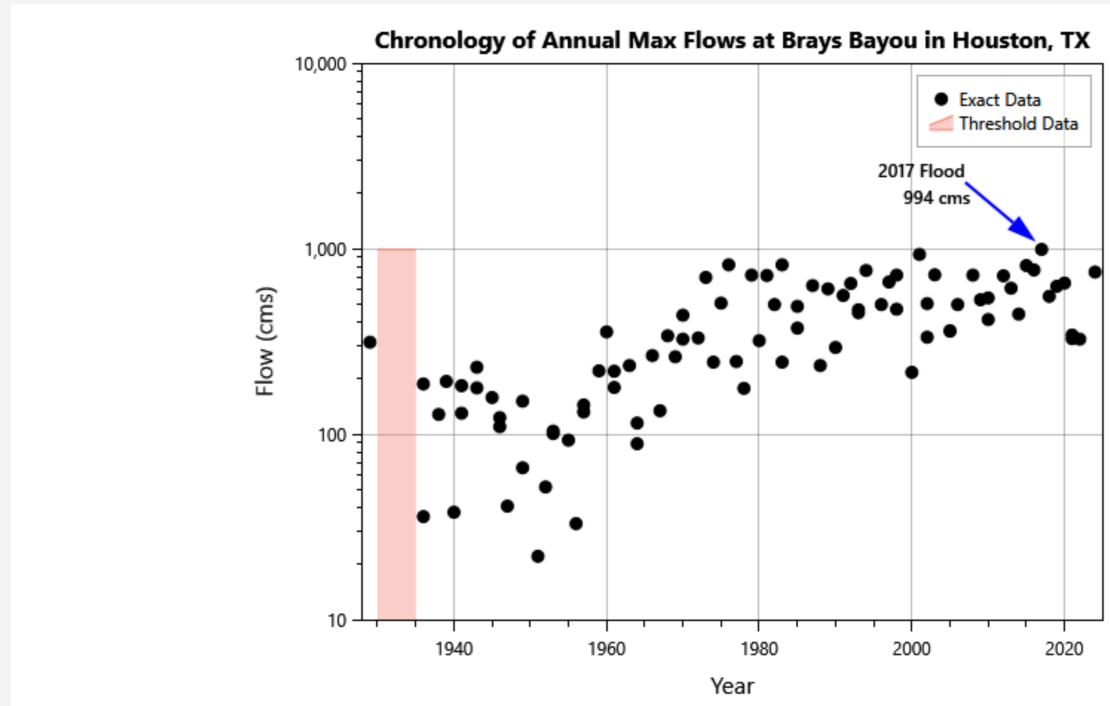
The model

Choosing a model

Results

Discussion

THE RECORD IS A MODELING CHOICE



Annual maximum instantaneous flow at USGS 08075000, 1929-2024; the shaded band is 1930-1935, when only a perception threshold is known (Smith et al., 2026, Fig. 2, CC BY 4.0).

THE LIKELIHOOD

The stationary likelihood (Smith et al., 2026, eq. 1):

$$L(\mathbf{x} \mid \theta) = \prod_{i=1}^N f(x_i \mid \theta)$$

In the **nonstationary model** the parameters depend on the year, θ_t (Smith et al., 2026, eq. 2):

$$L(\mathbf{x} \mid \theta) = \prod_{t=1}^T f(x_t \mid \theta_t)$$

A PERCEPTION THRESHOLD IS A CENSORED OBSERVATION

A perception threshold is “the flow above which an event would have been recorded in any given year” (Smith et al., 2026, p. 11, eq. 3).

$$L(\mathbf{x} \mid \theta) = \prod_{t=1}^T \begin{cases} f(x_t \mid \theta_t) & \text{exact data} \\ F(x_t^U \mid \theta_t) & \text{left censored (below threshold)} \\ 1 - F(x_t^L \mid \theta_t) & \text{right censored (above threshold)} \\ F(x_t^U \mid \theta_t) - F(x_t^L \mid \theta_t) & \text{interval censored} \end{cases}$$

f is the probability density and F the cumulative distribution function.

BRAYS BAYOU LIKELIHOOD

- 1929 and 1936-2024: $f(x_t \mid \theta_t)$.
- Each of 1930-1935: $F(994 \mid \theta_t)$.

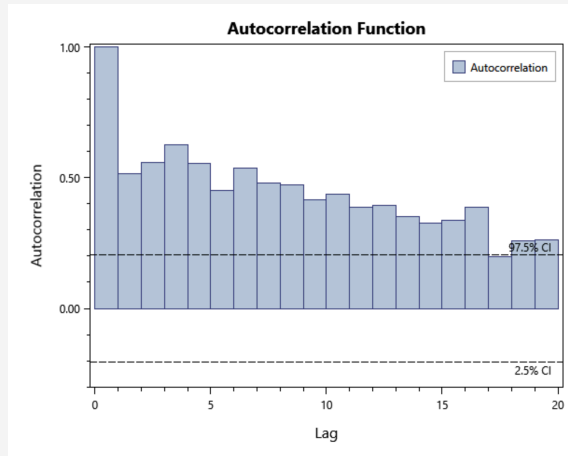
A year with no measured flood still carries information: six years in a row below 994 m³/s rules out parameter values that make 994 m³/s common.

For more, read Gelman et al. (2014), §8.7, on censored data in a Bayesian model.

Subperiod tests compare 1929-1980 against 1981-2024 (Smith et al., 2026, p. 13, Table 2).

Test	Null hypothesis	p -value
Jarque-Bera	Normality	0.0015
Ljung-Box Q	No autocorrelation	$< 10^{-15}$
Wald-Wolfowitz	Independence and stationarity	1.84×10^{-10}
Mann-Whitney U	Homogeneity	4.55×10^{-11}
Mann-Kendall	No monotonic trend	1.33×10^{-13}
Linear trend	$\beta_1 = 0$	$< 10^{-15}$
Equal-variance t -test	$\mu_1 = \mu_2$	4.23×10^{-13}
Unequal-variance t -test	$\mu_1 = \mu_2$	5.91×10^{-12}
F -test	$\sigma_1^2 = \sigma_2^2$	8.76×10^{-7}
Mixture model	Unimodality	0.0004

THE AUTOCORRELATION OF A TRENDING RECORD



Autocorrelation of the Brays annual maxima out to lag 20, with 95% bounds (Smith et al., 2026, Fig. 3, CC BY 4.0).

- Nearly every lag out to 20 sits above the bound.
- A trend does this: early years are all low and late years all high, so any two years close in time look alike.
- The paper reads it as “characteristic of a strong monotonic trend” (Smith et al., 2026, p. 13).

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LOG-PEARSON TYPE III

LP3 is used “consistent with federal guidelines established in Bulletin 17C” (Smith et al., 2026, p. 8).

- Like the GEV, it is fit to annual maxima.
- “Log” means it is fit to $\log_{10}$ of the flow.
- Think: log-normal with a skewness parameter added.
 - Positive means prone to floods far above the mean
 - Negative means the watershed caps the largest flow.
 - The skew is often similar at nearby gauges – we will revisit this next week!

TREND FORMULAS THE SOFTWARE OFFERS

Model	Formulation	Use case in the paper
Constant	$\mu_t = \alpha$	No detectable trend
Linear	$\mu_t = \alpha + \beta t$	Gradual urban expansion
Exponential	$\mu_t = \alpha e^{-\beta t}$	Trend slowing as a watershed stabilizes
Logistic	$\mu_t = \frac{\alpha}{1+e^{-\beta t}}$	Sprawl that plateaus
Power	$\mu_t = \alpha t^\beta$	Accelerating imperviousness
Quadratic	$\mu_t = \alpha + \beta t + \gamma t^2$	Rise then plateau
Cubic	$\mu_t = \alpha + \beta t + \gamma t^2 + \delta t^3$	Sequential land-use changes
Sinusoidal	$\mu_t = \alpha + \beta \sin(2\pi\gamma t + \delta)$	Multidecadal climate cycles
Step	$\mu_t = \mu_1$ for $t \leq t_c$, μ_2 for $t > t_c$	A land-cover change in a known year

TIME IS THE COVARIATE

💡 Think

Monday's recipe said to choose a covariate by mechanism. What does *t stand in* for at Brays?

- Imperviousness and channelization, neither of which is in the model.
- The authors say so: “formal covariate-based attribution is beyond the current scope” (Smith et al., 2026, p. 13).

BAYES' THEOREM IN SMITH'S NOTATION

Smith et al. (2026), eq. 4:

$$p(\theta \mid \mathbf{x}) = \frac{L(\mathbf{x} \mid \theta) \cdot \pi(\theta)}{\int L(\mathbf{x} \mid \theta) \cdot \pi(\theta) d\theta}$$

The joint prior over p parameters, then with the parameters independent (Smith et al., 2026, eqs. 5-6):

$$\pi(\theta) = \pi(\theta_1 \cap \theta_2 \cap \cdots \cap \theta_p)$$

$$\pi(\theta) = \prod_{i=1}^p f(\theta_i)$$

THE DEFAULT PRIORS

- Flat priors on location, scale, and shape, “centered near the likelihood with broad spread” (Smith et al., 2026, p. 5).
- A Jeffreys prior on the scale, $\pi(\sigma) \propto 1/\sigma$, “equivalent to using a log-link function” (p. 5).
- That is a flat prior on $\log \sigma$: the same log scale Monday’s model used for σ .
- Trend parameters (slopes, change points) get uninformative priors, and priors on the other parameters apply at the final year T (p. 6).

PRIOR INFORMATION ON A FLOOD QUANTILE

A quantile prior puts a distribution on the flood x_p with annual exceedance probability p , with hyperparameters θ_h (Smith et al., 2026, eq. 7):

$$h(x_p) = f(x_p \mid \theta_h)$$

The quantile is a function of the parameters in the final year (Smith et al., 2026, eq. 8):

$$x_p = F^{-1}(p \mid \theta_T)$$

A QUANTILE PRIOR BECOMES A PRIOR ON THE PARAMETERS

Substitute to evaluate it in parameter space (Smith et al., 2026, eq. 9):

$$\pi_q(\theta) = h(F^{-1}(p \mid \theta_T))$$

The posterior multiplies in every source (Smith et al., 2026, eq. 10):

$$p(\theta \mid \mathbf{x}) \propto L(\mathbf{x} \mid \theta) \cdot \pi(\theta) \cdot \pi_q(\theta)$$

“The quantile prior acts as a penalty function, rewarding parameter sets that produce flood quantiles consistent with the prior information” (p. 6). O.C. Fisher uses one, taken from a rainfall-runoff model; Brays does not.

SAMPLING THE POSTERIOR

RMC-BestFit samples with DE-MCzs, a differential-evolution Markov chain with a “snooker update” (Smith et al., 2026, p. 9).

- It uses no gradients.
- It scales poorly to many parameters, where the NUTS sampler you used in `Turing` scales well (by leveraging gradients)
- It can do better on a posterior with several modes, or a likelihood with discrete parameters, such as the step model’s change year t_c .

THE POSTERIOR PREDICTIVE DISTRIBUTION

The full distribution of a flood k years into the future (Smith et al., 2026, eq. 14):

$$p(x_{T+k} \mid \mathbf{x}) = \int f(x_{T+k} \mid \theta_{T+k}) \cdot p(\theta \mid \mathbf{x}) d\theta$$

- It “accounts for both parameter uncertainty (epistemic) and natural variability (aleatory)” (p. 10), as in labs 4 and 5.
 - This is a *narrow* definition of epistemic uncertainty
- Think of it as “averaging over the posterior”

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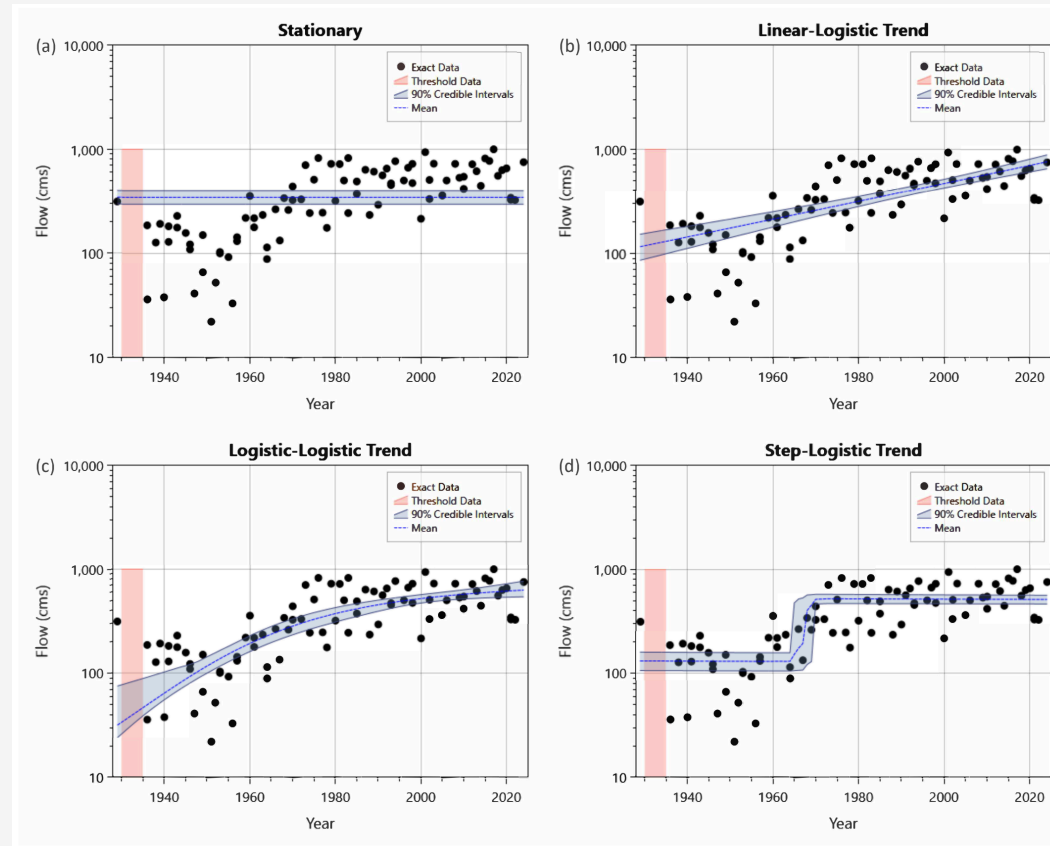
Discussion

SEVEN CANDIDATE MODELS FOR BRAYS

Model comparison (Smith et al., 2026, p. 15, Table 3), with AIC, BIC, and DIC revisited in week 8.

Model	k	AIC	BIC	DIC	Δ DIC
Stationary	3	1233.73	1241.23	1228.36	73.06
Linear (μ)	4	1182.14	1192.14	1180.76	25.45
Logistic (μ)	4	1179.94	1189.94	1179.23	23.93
Step (μ)	5	1177.04	1189.54	1162.28	6.98
Linear-logistic (μ - σ)	5	1168.70	1181.20	1172.04	16.74
Logistic-logistic (μ - σ)	5	1161.46	1173.96	1160.26	4.95
Step-logistic (μ - σ)	6	1165.24	1180.24	1155.30	0.00

FOUR PLAUSIBLE MODELS



The 2-year return level under (a) the stationary, (b) linear-logistic, (c) logistic-logistic, and (d) step-logistic models, with 90% credible intervals (Smith et al., 2026, Fig. 4, CC BY 4.0).

THE FITTED CHANGE POINT AND THE VARIANCE

- The change point is estimated at 1968, near the start of Houston's most intensive development in the early 1970s (Smith et al., 2026, p. 14).
- The location jumps at the change point, while the scale, the spread of the log-flows, declines smoothly over the whole record with no jump (logistic, $\beta = -0.014$).
- Early floods spanned about 20-300 m³/s, more than a factor of ten; after urbanization, about 200-800 m³/s, a factor of four (p. 14).

WHY URBANIZATION NARROWS THE LOG-SPACE SPREAD

“In a pre-urbanization watershed, flood magnitudes are highly variable because antecedent soil moisture, infiltration capacity, and vegetation interception modulate rainfall-runoff partitioning differently for each event” (Smith et al., 2026, p. 23).

Pave it over and the response “approaches a near-deterministic conversion of rainfall to runoff” (p. 24).

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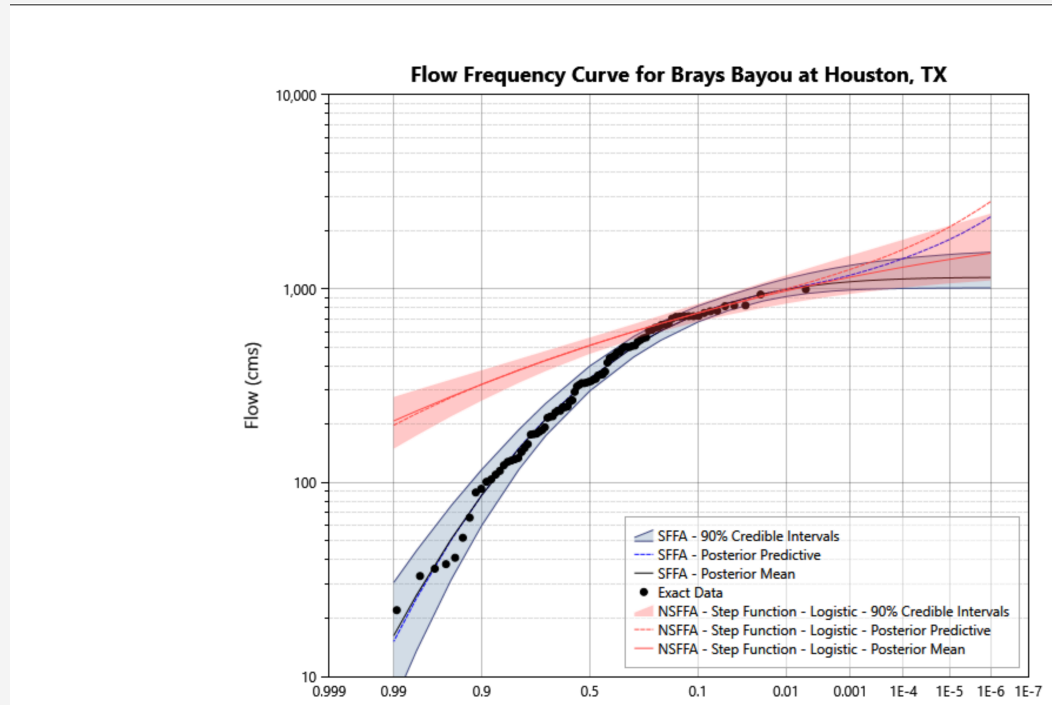
Discussion

STATIONARY AND NONSTATIONARY FLOOD QUANTILES

Posterior predictive estimates and 90% intervals in m³/s, the nonstationary column at 2024 conditions (Smith et al., 2026, p. 16, Table 4).

AEP (return period)	Stationary	90% CI	Nonstationary	90% CI
0.50 (2-yr)	346	[297, 400]	511	[462, 560]
0.20 (5-yr)	607	[542, 673]	661	[603, 727]
0.10 (10-yr)	744	[674, 815]	749	[675, 836]
0.02 (50-yr)	940	[865, 1046]	924	[798, 1076]
0.01 (100-yr)	993	[912, 1126]	996	[838, 1172]

FREQUENCY CURVES MEET IN THE TAIL



Stationary (blue) and nonstationary (red) frequency curves for Brays Bayou, with 90% intervals; the plotting positions assume stationarity (Smith et al., 2026, Fig. 5, CC BY 4.0).

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STEP MODEL

Think

The selected model has one level before 1968 and one after, forever. What would violate that?

- More urbanization after 1968.
- More infrastructure: channelization and detention basins.

HOW YOU MIGHT ADDRESS IT

- A better covariate: impervious area fraction in place of the year, which the paper names as future work (Smith et al., 2026, p. 25).
- A different method: fit the extreme value model to rainfall, and run it through a flood model that represents land use and infrastructure.
- The paper's own list adds hierarchical pooling across sites, model averaging over the trend forms, and alternative distributions (pp. 24-25).